

Large Language Models for Clinical Diagnosis and Treatment Recommendation

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Abstract

Clinical decision support systems powered by large language models represent a promising frontier in medical artificial intelligence. This paper presents Project Omega, a systematic comparative evaluation of Meta's LLaMA-3.3-70B and Google's Gemma-4-31B on the task of clinical diagnosis hypothesization and treatment recommendation. Using a cohort of 2,000 patient admissions from the MIMIC-IV clinical database, we construct structured prompts from electronic health records and evaluate model outputs across output verbosity, drug overlap, BLEU and ROUGE-1/2/L scores, and four clinical quality dimensions (accuracy, completeness, safety, usefulness) assessed by an autonomous LLM evaluator (GPT-OSS 120B). We additionally conduct a temperature sensitivity experiment on 1,000 patients and apply LIME-based explainability analysis. Gemma-4-31B wins 97.9% of head-to-head evaluations while simultaneously scoring lower on BLEU and ROUGE, revealing a fundamental limitation of n-gram metrics for clinical text generation tasks.

1. Introduction

The integration of artificial intelligence into clinical workflows has long been a central ambition of biomedical informatics. Electronic health records contain rich, longitudinal data spanning diagnoses, laboratory measurements, vital signs, pharmacological histories, and procedural notes. Transforming this data into actionable clinical recommendations is a task that historically required years of specialist training, yet represents a natural target for modern language models trained on vast biomedical corpora [15].

Large language models have demonstrated remarkable capabilities on structured medical benchmarks. GPT-4 achieves passing scores on the United States Medical Licensing Examination [10], and biomedical variants such as BioBERT [7] and ClinicalBERT [1] have established strong baselines on clinical information extraction tasks. However, standardized multiple-choice examinations do not capture the open-ended complexity of real clinical reasoning: generating a differential diagnosis list, proposing a treatment plan, identifying drug interactions, and communicating safety concerns to a clinical practitioner.

The rapid proliferation of open-weight language models has made it feasible to deploy powerful inference pipelines without proprietary APIs. Meta's LLaMA family [3] and Google's Gemma models [2] now offer strong instruction-following capabilities at scales accessible to academic research groups. Yet systematic comparative studies of these models on realistic clinical tasks grounded in real patient data remain scarce [5, 14].

This paper addresses this gap through Project Omega, a pipeline-based evaluation framework built on the MIMIC-IV clinical database [6]. Our contributions are:

- **Pipeline Design.** A scalable Python pipeline processes structured EHR data from a DuckDB-indexed MIMIC-IV database, constructs multi-section clinical prompts, and queries two LLMs across 2,000 patient admissions.
- **Multi-Dimensional Evaluation.** Outputs are evaluated

across output verbosity, drug overlap ratios, BLEU and ROUGE-1/2/L scores, and four clinical quality axes assessed by an autonomous GPT-OSS 120B evaluator [16].

- **Temperature Sensitivity.** The effect of the generation temperature on Gemma-4-31B output quality is investigated across 1,000 patients at three settings.
- **Explainability.** LIME [13] is applied to identify which prompt sections most influence each model's output behavior.

Our central finding is a counterintuitive reversal of rankings between lexical similarity metrics and clinical quality assessments. LLaMA-3.3-70B achieves higher BLEU and ROUGE scores due to its concise outputs, yet Gemma-4-31B wins 97.9% of head-to-head clinical evaluations. This result has direct methodological implications for the evaluation of clinical text generation systems.

2. Related Work

2.1. NLP in Clinical Medicine

Natural language processing has been applied to clinical text for decades, beginning with rule-based systems for named entity recognition in discharge summaries. The introduction of transformer architectures fundamentally transformed the field. BioBERT [7] and ClinicalBERT [1] demonstrated that domain-adaptive pretraining on biomedical and clinical corpora yields substantial gains on downstream clinical tasks including NER, relation extraction, and question answering. The digitization of healthcare has generated large Electronic Health Records (EHR) repositories, which serve as crucial training and evaluation data for clinical machine learning systems. Although closed-source models initially dominated early clinical feasibility studies, privacy constraints and data sovereignty regulations in the healthcare sector have led to growing interest in open-weight alternatives, which offer the possibility of local deployment ensuring strict compliance with patient data privacy.

2.2. LLMs on Medical Tasks

Singhal et al. [15] show that instruction-tuned language models encode substantial clinical knowledge and achieve competitive performance on medical question-answering. Nori et al. [10] demonstrate GPT-4’s ability to pass standardized medical licensing examinations, while Jin et al. [4] present large-scale question-answering datasets demonstrating the gap between model capabilities and expert human performance. Performance on free-form clinical generation situated within real patient records, however, remains considerably less studied. Our work directly targets this gap by constructing prompts from structured EHR data rather than from curated examination questions.

2.3. LLM-as-Judge Evaluation

The LLM-as-Judge paradigm, formalized by Zheng et al. [16] through the MT-Bench framework, proposes using a capable language model as an autonomous preference evaluator. The authors demonstrate strong agreement between GPT-4 judgments and human annotations across open-ended tasks. We adopt this paradigm using GPT-OSS 120B as our evaluator, scoring model outputs on four clinical quality dimensions across all 2,000 patients.

2.4. Evaluation Metrics for Text Generation

BLEU [11] and ROUGE [8] remain the dominant automatic metrics for text generation evaluation. Both measure lexical n-gram overlap between generated text and a reference string. Their applicability to clinical text is contested: a comprehensive clinical recommendation including treatment rationales, safety warnings, and procedural guidance will naturally show lower BLEU and ROUGE scores relative to a terse output, even if it is clinically superior. Our results empirically confirm this limitation [9].

3. Methodology

3.1. Dataset and Patient Cohort

We use the MIMIC-IV clinical database [6], a freely accessible collection of de-identified health records from over 40,000 patients admitted to the critical care units of Beth Israel Deaconess Medical Center between 2008 and 2024, locally indexed using DuckDB.

Patient eligibility requires at least one ICD diagnosis code, at least one laboratory event, and at least one prescription record. From the eligible pool, a reproducible random cohort of 2,000 patients is drawn using a seeded shuffle (seed = 42). Cohort identifiers are persisted to CSV to guarantee identical patient selection across pipeline restarts.

3.2. Two-Phase Data Loading

To maintain a constant memory footprint, we adopt a two-phase architecture. Phase 1 retrieves patient identifiers, demographics, and all ICD-coded diagnoses in a lightweight query held in memory throughout the run. Phase 2 loads, for each batch of 50 patients, laboratory results with human-readable labels, prescriptions, ICU stay records, chartevents vitals (heart rate, SpO₂, temperature, blood pressure, respiratory rate), OMR anthropometrics, ICD-coded procedures, and microbiology culture results. Batch data is released after each iteration.

3.3. Prompt Construction

Clinical data for each patient is assembled into a structured, eight-section prompt: Patient Demographics, Diagnoses (primary flagged separately), Laboratory Results (up to 15 most recent numeric measurements), ICU Vitals (up to 12 chartevents measurements), Vital Signs and Anthropometrics (OMR records), Procedures Performed, Microbiology and Culture Results, and ICU Stay duration.

The task instruction requires each model to respond in exactly four structured sections: (1) Additional Diagnosis Hypotheses, listing 3 to 5 differential or comorbid diagnoses with clinical rationales; (2) Recommended Treatments with drug names and typical doses; (3) Safety Flags and Drug Interactions; and (4) Confidence Level with a one-sentence explanation.

3.4. Models

LLaMA-3.3-70B-Versatile is Meta’s 70-billion-parameter decoder-only transformer [3], accessed via the Groq inference API. **Gemma-4-31B-IT** is Google’s 31-billion-parameter instruction-tuned model [2], accessed via the Google Generative AI API. Both models use temperature $T = 0.3$ and a maximum of 2,500 output tokens. The autonomous evaluator **GPT-OSS 120B** is queried at temperature $T = 0.1$ with up to 3,000 tokens.

3.5. Drug Overlap Analysis

True prescribed medications are retrieved from the MIMIC-IV prescriptions table. A fuzzy string matching algorithm (Python difflib SequenceMatcher, threshold ≥ 0.75) compares drug tokens extracted from model outputs against the ground-truth prescription list, yielding absolute match counts and recall ratios normalized by the number of true prescriptions.

3.6. Temperature Sensitivity Experiment

Gemma-4-31B is re-queried on a subset of 1,000 patients at three temperature settings $T \in \{0.1, 0.5, 1.0\}$. BLEU and ROUGE-1 scores are computed against the patient’s concatenated ICD diagnosis descriptions, producing 3,000 evaluation pairs.

4. Evaluation Strategy

4.1. Lexical Similarity

BLEU [11] is computed using NLTK’s `sentence_bleu` with Chen and Cherry smoothing (Method 1). ROUGE-1, ROUGE-2, and ROUGE-L [8] are computed using the `rouge-score` library. Reference text is the concatenated ICD long-title diagnosis string for each admission.

4.2. LLM-as-Judge

GPT-OSS 120B scores each pair of model outputs on four axes, each from 1 to 25 (total out of 100):

- **Clinical Accuracy.** Correctness of differential diagnoses, recommended drugs, and dosing.
- **Completeness.** Coverage of all clinically relevant aspects without omitting critical information.
- **Safety.** Identification of drug interactions, contraindications, and safety warnings.

- **Practical Usefulness.** Clarity, structure, and actionability for a clinical practitioner.

The evaluator additionally designates a head-to-head winner (model1, model2, or tie) and provides a short comparison rationale returned as structured JSON parsed programmatically.

5. Results and Analysis

Figure 1 presents the full evaluation dashboard. Table 1 summarizes quantitative results.

Table 1: Comparative results ($n = 2,000$). BLEU and ROUGE values are medians. Evaluator scores are medians as percentages of maximum.

Metric	LLaMA-70B	Gemma-31B
Output length (chars)	2,100	4,500
Drug overlap (count)	1.9	2.1
Drug overlap (% true Rx)	3.8%	4.2%
BLEU	0.052	0.018
ROUGE-1	0.220	0.121
ROUGE-2	0.071	0.040
ROUGE-L	0.135	0.105
Evaluator total (%)	50	75
Accuracy (%)	65	85
Safety (%)	56	85
Completeness (%)	48	80
Usefulness (%)	48	82
Head-to-head wins	38 (1.9%)	1,959 (97.9%)

5.1. Output Verbosity

A pronounced difference in output verbosity characterizes the two models. LLaMA-3.3-70B produces consistently concise outputs (median $\approx 2,100$ characters, IQR 1,800-2,300), while Gemma-4-31B generates substantially longer responses (median $\approx 4,500$ characters, IQR 3,500-5,000). This $2.1\times$ difference in output length is the primary driver of the divergence in n-gram similarity scores discussed below.

5.2. Drug Overlap

Both models achieve modest drug overlap counts (median 2 matched drugs per patient), reflecting the inherent difficulty of exact pharmacological recommendation from ICD diagnosis codes alone. Drug overlap percentage remains below 10% for both models, consistent with findings in related LLM pharmacology studies. Gemma’s wider output distribution yields a marginally higher count at the upper quartile, suggesting its more comprehensive outputs occasionally recover additional relevant medications.

5.3. Lexical Similarity and the Brevity Effect

LLaMA-3.3-70B achieves higher median BLEU (0.052 vs. 0.018) and ROUGE-1 (0.220 vs. 0.121) scores than Gemma-4-31B. The explanation lies in the *brevity effect*: LLaMA’s shorter, diagnosis-focused outputs contain a higher proportion of ICD-related vocabulary, yielding greater lexical overlap with the ICD reference strings.

Gemma’s comprehensive outputs, which include treatment plans, drug interaction warnings, and procedural guidance, contain proportionally less diagnosis-specific vocabulary and are penalized by n-gram precision metrics.

This result has a direct methodological implication: BLEU and ROUGE should not serve as primary quality indicators for open-ended clinical generation tasks where reference strings are concise ICD descriptions.

5.4. Clinical Quality Evaluation

The GPT-OSS 120B evaluator reveals a decisive advantage for Gemma-4-31B across all four quality dimensions. Gemma achieves a median total score of approximately 75% versus LLaMA’s 50%. In head-to-head comparisons, Gemma wins 1,959 of 2,000 evaluations (97.9%), LLaMA wins 38 (1.9%), and 3 result in a tie.

The largest performance gaps appear in Completeness (Gemma 80% vs. LLaMA 48%) and Usefulness (Gemma 82% vs. LLaMA 48%), confirming that Gemma’s verbosity translates into genuine clinical value. Safety scores differ substantially (Gemma 85% vs. LLaMA 56%), suggesting LLaMA more frequently omits drug interaction warnings and contraindication notices.

5.5. Temperature Sensitivity

Table 2 shows temperature’s effect on Gemma output quality. BLEU scores remain uniformly low (≈ 0.010 - 0.011) across all three temperatures. ROUGE-1 shows a slight upward trend from $T = 0.1$ (median ≈ 0.100) to $T \in \{0.5, 1.0\}$ (median ≈ 0.122), suggesting that higher temperatures introduce vocabulary diversity that marginally increases recall. Overall, Gemma-4-31B is largely robust to this hyperparameter within the tested range.

Table 2: Temperature sensitivity for Gemma-4-31B ($n = 1,000$ per setting). Values are medians.

Temperature	BLEU	ROUGE-1
0.1	0.010	0.100
0.5	0.011	0.122
1.0	0.011	0.123

6. Explainability Analysis

6.1. LIME Setup

To understand which sections of the clinical prompt most influence each model’s response, we apply LIME [13] to 50 randomly sampled patients per model. For each patient, 800 perturbed prompt variants are generated by randomly masking token subsets. Each perturbed prompt is submitted to the model and the resulting output is compared to the original output using TF-IDF cosine similarity. Let Y_{orig} represent the original, unperturbed model output for a given prompt. When LIME perturbs a clinical prompt X to generate a subset prompt X_{pert} , a local vectorizer converts the text into TF-IDF space [12]. The proxy scoring function computes cosine similarity between the perturbed text vector and the reference vector of Y_{orig} :

$$\text{Similarity}(X_{\text{pert}}, Y_{\text{orig}}) = \frac{\mathbf{V}(X_{\text{pert}}) \cdot \mathbf{V}(Y_{\text{orig}})}{\|\mathbf{V}(X_{\text{pert}})\| \cdot \|\mathbf{V}(Y_{\text{orig}})\|} \quad (1)$$

Tokens whose masking drastically drops the cosine similarity to the original output are assigned positive LIME weights, pinpointing them as influential drivers of the model’s conclusion. A linear surrogate model fitted to these input-output pairs identifies the most influential tokens.

An important methodological note: this setup measures *lexical influence*, specifically whether a token’s presence causes the model to produce vocabulary similar to its unperturbed output. This constitutes a lexical-overlap proxy for explainability rather than a measure of semantic influence or causal reasoning.

6.2. Results

Figure 2 displays the top globally influential words aggregated across all sampled patients for each model. For both LLaMA-3.3-70B and Gemma-4-31B, tokens from the Diagnoses and Laboratory Results sections consistently rank among the most influential. This confirms that both models anchor primarily on clinical diagnostic context rather than on demographic information or procedural history.

Section-level heatmap analysis reveals a notable difference: LLaMA’s outputs are most strongly influenced by primary diagnosis terminology, reflecting its concise, diagnosis-centric generation style. Gemma exhibits more distributed influence across the Diagnoses, Laboratory Results, and Vital Signs sections, consistent with its broader, more integrative output behavior.

7. Discussion and Limitations

7.1. Discussion

Three key findings emerge from this study. First, the brevity effect demonstrates that standard automatic metrics can produce misleading model rankings for clinical text generation. Shorter outputs that closely mirror reference diagnosis strings score higher on BLEU and ROUGE, despite being clinically inferior. This result advocates for multi-dimensional LLM-based evaluation as the standard for clinical text generation benchmarks.

Second, Gemma-4-31B’s comprehensive output style, characterized by detailed treatment justifications, explicit safety warnings, and structured confidence ratings, is strongly preferred by an autonomous clinical evaluator across all four quality dimensions. The 97.9% head-to-head win rate across 2,000 patients is a robust result that substantially reduces the influence of prompt-level variance.

Third, the temperature sensitivity experiment demonstrates that Gemma-4-31B produces consistently quality outputs across $T \in \{0.1, 0.5, 1.0\}$, with only marginal variation in lexical diversity metrics. This robustness is practically significant for clinical deployment pipelines operating under fixed inference configurations.

7.2. AI-by-AI Evaluation Validity

The use of GPT-OSS-120B as an autonomous clinical judge introduces a potential *self-reinforcement bias*: a judge model may award high scores to clinically incorrect yet fluent and persuasive hypotheses simply because they are well-structured and verbose. This risk is mitigated in our framework by combining qualitative LLM judgment with

a deterministic drug-verification module. The fuzzy string matching protocol (difflib SequenceMatcher, threshold ≥ 0.75) cross-checks every recommended drug against the actual hospital pharmacy records stored in the MIMIC-IV prescriptions table, grounding the evaluator’s qualitative preference in objective pharmacological reality.

7.3. Hallucination Containment

In the ICU setting, hallucinations can be classified as *intrinsic* (the model misinterprets a laboratory value within the record) or *extrinsic* (the model proposes a treatment or diagnosis that never appears in the patient’s EHR). We constrain token selection to a low-entropy distribution by fixing the generation temperature at $T = 0.3$, which reduces the probability of sampling rare or unsupported tokens. Additionally, requiring a one-sentence clinical justification for every additional diagnosis makes the outputs easier to audit against the source EHR documents, thereby containing both intrinsic and extrinsic hallucinations.

7.4. Limitations

The LIME explainability analysis employs a lexical-overlap proxy rather than true semantic attribution. Future work should apply gradient-based attribution methods for more rigorous interpretability.

Drug overlap relies on fuzzy string matching, which may both over-count matches between pharmacologically unrelated drugs sharing similar character sequences, and under-count paraphrased or abbreviated drug names. A standardized pharmacological ontology such as RxNorm would yield more precise recall measurements.

The autonomous evaluator may introduce preference biases, potentially favoring well-structured, verbose outputs independent of actual clinical correctness. Human expert validation by practicing clinicians would substantially strengthen evaluation validity.

Finally, our cohort represents a single institution and a historical time period (2008-2024). Generalizability to contemporary clinical settings and diverse healthcare systems requires validation on more recent and multi-institutional datasets.

8. Conclusion

This paper presents Project Omega, a comprehensive evaluation of LLaMA-3.3-70B and Gemma-4-31B for clinical diagnosis hypothesization and treatment recommendation on 2,000 MIMIC-IV patient admissions. Our pipeline combines drug overlap analysis, lexical similarity metrics, LLM-based clinical quality assessment, temperature sensitivity experiments, and LIME explainability analysis into a reproducible evaluation framework.

Gemma-4-31B substantially outperforms LLaMA-3.3-70B in clinical quality, winning 97.9% of head-to-head evaluations, while simultaneously scoring lower on BLEU and ROUGE. This contradiction resolves when output verbosity is considered: Gemma's longer, more comprehensive outputs are penalized by n-gram metrics that reward lexical conciseness relative to short ICD reference strings. This finding has direct implications for how clinical text generation systems should be evaluated, and motivates the adoption of multi-dimensional LLM-based evaluation frameworks alongside traditional metrics.

Future work should address the following directions:

- **Validation by practicing clinicians:** Integrating practicing medical clinicians to perform a blind review of LLM outputs, rigorously cross-verifying the alignment correlation between the AI judge and human experts.
- **Evaluation on contemporary multi-institutional datasets:** Testing generalizability across diverse healthcare systems and more recent data collections.
- **Fine-tuning on clinical corpora:** Domain-adaptive training to improve diagnostic accuracy and reduce hallucination rates.
- **Targeted assessment of safety-critical subpopulations:** Focused evaluation on pediatric, oncological, and other high-risk patient groups.
- **Loop in human clinicians:** Integrating practicing medical clinicians to perform a blind review of LLM outputs, rigorously cross-verifying the alignment correlation between the AI judge and human experts.
- **Vector extensions with images:** Expanding the ingestion pipeline beyond structured alphanumeric data to integrate vector embeddings of raw medical imaging transcripts (radiology reports).

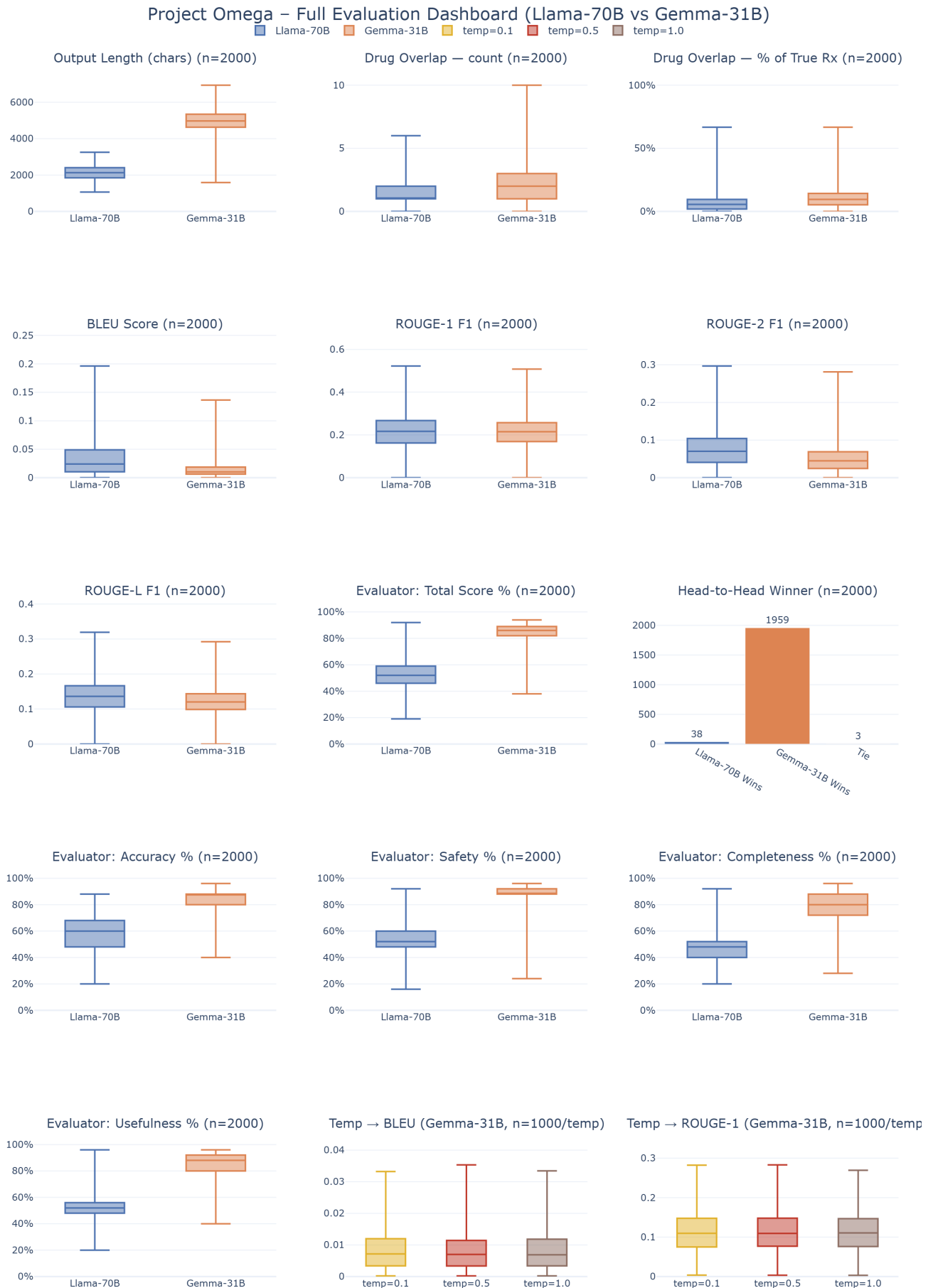


Figure 1: Full evaluation dashboard for 2,000 MIMIC-IV patients. Rows 1–2: output statistics and lexical similarity metrics. Row 3: head-to-head winner counts and evaluator total scores. Rows 4–5: evaluator subscores and temperature sensitivity analysis (Gemma-4-31B, $n = 1,000$ per temperature).

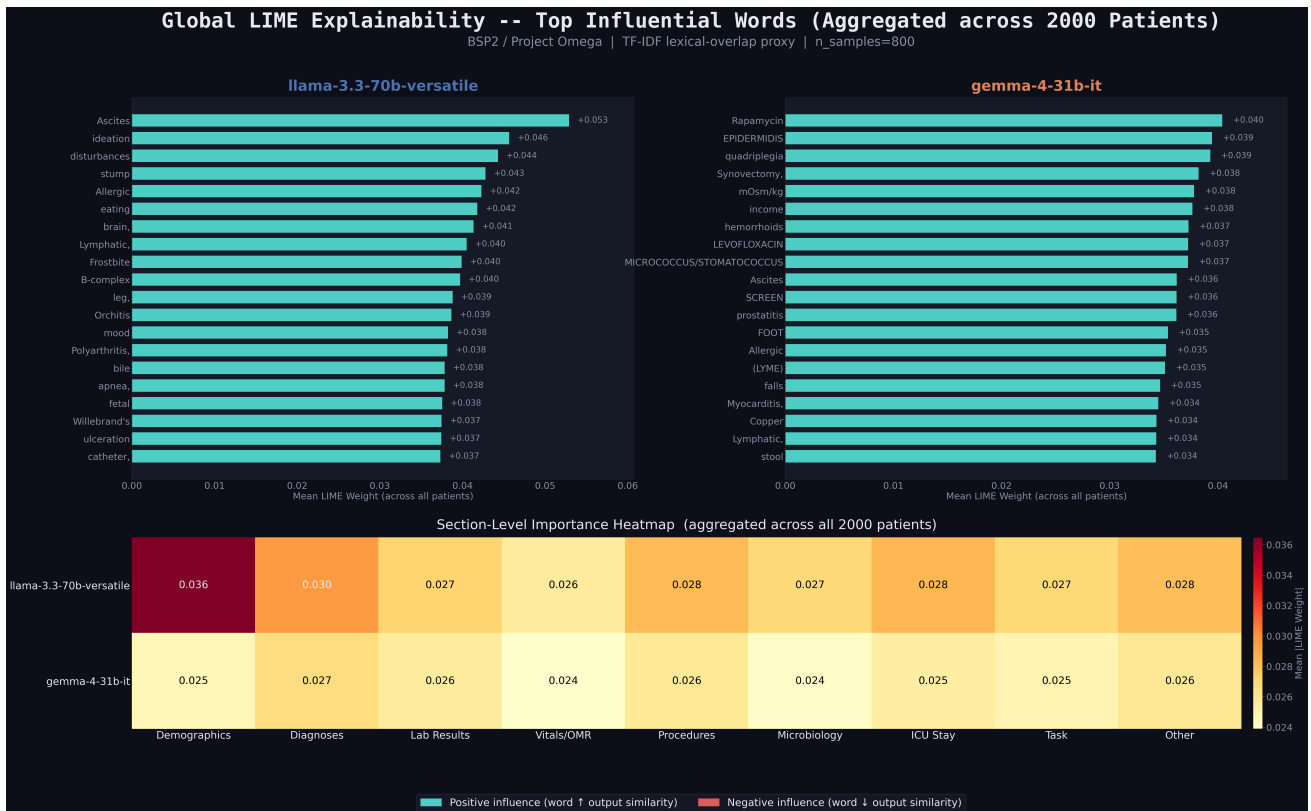


Figure 2: Global LIME explainability results for LLaMA-3.3-70B (left) and Gemma-4-31B (right), aggregated across 50 patients each. Bars show mean absolute influence weight; color indicates direction (teal: positive, coral: negative). Bottom panels show section-level importance heatmaps.

Work Split

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- **Exploratory Research:** Conducted initial research on MIMIC-IV dataset functions and architecture, drafting a brief used to align the team's technical scope.
- **Review of Academic Literature:** Researched and analyzed similar medical LLM studies to establish the comparative framework for the final report.
- **Project Infrastructure:** Initialized the team's GitHub repository and organized the foundational documentation to establish a collaborative workflow.
- **Script Prototyping:** Developed the early-stage Python script `preprocess_mimic.py` to clean and parse the MIMIC-IV demo dataset, allowing the team to begin preliminary testing while awaiting access to the full dataset.
- **Algorithmic Explorations (Unused Features):** Programmed experimental features, including an image processing module (`images_mimic_v1.py`) to include images in the table of relevant data, `pipelines-images-brainstorm.py` to add image processing to the pipeline, and a patient readmission function `readmissions.py`, which served as valuable early proof-of-concepts despite being scoped out of the final model due to time constraints.
- **Code Review and Support:** Acted as a peer-reviewer by auditing team-written code to ensure logical correctness. Served as a technical resource within the group, explaining important code segments to keep all team members aligned.
- **Translation:** Manually translated the original English abstract into French.
- **Writing:** Wrote Sections 2 and 3 of the report.
- **Formatting:** Properly formatted the final report and abstract.

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- **Designed Pipeline:** Built an automated system for API requests to ensure uninterrupted processing of 2,000 patients.
- **Integrated Models:** Integrated LLaMA-3.3-70B via multiple Groq API keys, and Gemma-4-31B via the Google Gemini API.
- **Integrated Evaluator Model:** Integrated GPT-OSS 120B via multiple API keys.
- **Implemented Error Handling:** Added robust catch mechanisms for API limits (429), exhausted credits (402), missing models (404), and permission denials (403), triggering automatic provider switching.
- **Hyperparameter Experiments:** Wrote `temperature_test.py` to use Gemma-4-31B and adjusted testing temperatures to $T \in \{0.1, 0.5, 1.0\}$.
- **Analysis Scripts:** Wrote `bleu_rouge_metrics.py`, `graph_maker.py`, and `lime_analyzer.py` to reflect the new pipeline architecture, model names, and chart labels.
- **Resume Functionality:** Implemented logic to track processed patients and seamlessly resume execution after interruptions.
- **Pipeline Execution and Data Collection:** Ran the end-to-end patient analysis pipeline, processing 2,000 cases to generate the final LLM response and evaluation datasets.
- **Data Extraction:** Ran all analytical scripts to extract final performance metrics, generation scores, and LIME explainability results.
- **Data Preprocessing and Local Database Setup:** Developed `init_db.py` to parse raw MIMIC-IV CSV files and initialize a high-performance local DuckDB database, enabling fast and reproducible extraction of the 2,000-patient cohort.
- **Prompt Engineering and EHR Data Structuring:** Designed structured prompt templates that dynamically map Electronic Health Records (ICD diagnoses, laboratory results, ICU vitals, and procedures) into cohesive, standardized inputs for the LLMs.
- **Writing and Presentation:** Wrote Section 1 (Introduction), Section 4 (Implementations and Results), Section 5 (Discussion and Limitations), and Section 6 (Conclusion) of the final report. Designed and compiled the final project presentation slides.
- **Version Control and Deployment:** Uploaded and managed the codebase and project files on GitHub for version control and collaborative access.

AI Usage Disclosure

Generative AI tool (Gemini Pro 3.1) were used to assist with code drafting, debugging, and editing of written content throughout this project. All architectural decisions, methodology design, model selection, and final content review remain the responsibility of the authors.

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